

CAREER: Foundations of Cost-Aware Energy Characterization and Optimization for Computing Systems

Overview

Over the past decade, AI and high-performance computing have driven rapid growth in energy-intensive processors, accelerators, memory, storage, and datacenter platforms. As these systems scale, energy has become not only a resource constraint, but also a major determinant of operating cost, hardware deployment, workload placement, and long-term system scalability. However, existing computer systems and architecture research still largely treats energy as a simple scalar quantity, such as power, joules, or energy-delay product. This abstraction is no longer sufficient for modern computing systems, where the real cost of energy depends on workload behavior, hardware architecture, utilization, lifecycle stage, deployment region, and time-varying operating conditions.

We envision a future where computer systems can reason about energy by its **cost-relevant properties**, rather than only by the amount consumed. To enable this future, it requires a holistic cross-layer approach that rethinks how systems characterize, attribute, and optimize energy across software and hardware. **The ultimate goal of this project is to introduce a new cost-aware energy characterization and optimization framework for computing systems, enabling high-performance, cost-efficient, and scalable system design.** The proposed research will be based on detailed characterization studies to understand how energy cost arises across modern AI and HPC systems. The insights gathered from these studies will be used to abstract new design principles, metrics, and optimization methods for cost-aware computing systems.

Our proposed cost-aware energy framework will comprise the following components: (a) **the modeling stack** will define new energy-cost abstractions that connect computing activities to cost drivers across hardware design, workload execution, deployment context, and lifecycle stage; (b) **the measurement stack** will quantify how energy cost varies across systems, workloads, and operating conditions, enabling fine-grained attribution of cost to hardware and software decisions; and (c) **the optimization stack** will integrate these cost-aware energy signals into architecture design space exploration, workload scheduling, and runtime adaptation. Together, these stacks make energy cost a first-class systems abstraction that can be quantified, attributed, and optimized across hardware and software decisions.

Intellectual Merit

This project advances computer systems and architecture by establishing energy cost as a first-class systems abstraction. Rather than treating energy as a uniform quantity, the project will reveal how real energy cost is shaped by hardware design, workload behavior, runtime decisions, deployment context, and lifecycle stage. This new foundation will enable computing systems to identify where energy costs originate, how they propagate across hardware and software layers, and when conventional energy metrics obscure important tradeoffs. The resulting insights will expose new optimization opportunities and enable systems that reduce real cost while maintaining performance and scalability.

Broader Impacts

This project will advance U.S. leadership in energy-efficient AI infrastructure and semiconductor innovation by helping computing systems reduce the real cost of energy use. The results can inform semiconductor design, HPC operation, hardware procurement, workload management, and regional deployment decisions, enabling more cost-efficient and scalable computing infrastructure. The project will also advance education and workforce development. New course modules, open-source tools, datasets, and hands-on projects will train students in cost-aware energy modeling and hardware-software optimization. Outreach through Purdue programs will introduce K–12 and undergraduate students to how computer systems and architecture design can shape the future of energy-efficient, scalable computing infrastructure.